

6(a)(i)	P – insulin ; Q – muscle / liver ; R – glycogen ;	3
6(a)(ii)	negative feedback ;	1

Question	Answer	Marks
6(b)	any seven from: 1 glucagon ; 2 binds to receptor on cell surface membrane (of liver cell) ; 3 G-protein activated ; 4 adenylyl cyclase activated ; 5 ATP converted to cAMP ; 6 second messenger ; 7 enzyme cascade / described ; 8 enzymes activated by phosphorylation ; 9 signal amplified ; 10 glycogen broken down into glucose / glycogenolysis ; 11 glucose enters blood ; 12 gluconeogenesis / described ;	7